AMERESCO
Green • Clean • Sustainable

Data Analyser ADVANCED

User guide for half-hourly energy data analysis



Where to start

Ranked by the estimated annual cost associated with closed-hours use above the reference: energy used when the building should be closed, beyond what never switches off. An associated cost, not a guaranteed saving. Meters with too little data to annualise are listed after, unranked. Click a heading to sort, or # to go back to the ranking.

Table to Excel

# ▲	Site	Days	Observed kWh	kWh/yr est.	£/yr est.	Base kW	Closed	Excess £/yr	Headroom	Data	Findings	
1	**Library (faulty meter)** Office / commercial · hours unconfirmed	364 99% cover	77,923	78,191	£19,548	2.3	36%	**£3,895**	-	• Poor	1 High 2 Medium 1 Low	Remove
2	**Town Hall** Office / commercial · 1200012345678 · hours unconfirmed	395 99% cover	87,931	81,312	£20,328	2.9	37%	**£3,311**	-	• Check	1 High 2 Medium	Remove
3	**St Marys Primary** Office / commercial · hours unconfirmed	364 99% cover	80,547	80,824	£20,206	2.9	36%	**£3,215**	-	• Check	1 High 2 Medium	Remove
4	**Riverside Leisure** Office / commercial · hours unconfirmed	363 99% cover	80,357	80,855	£20,214	2.9	36%	**£3,198**	-	• Check	1 High 2 Medium	Remove
-	**Part Day Site** Office / commercial · hours unconfirmed	60 99% cover	15,839	-	-	6.5	65%	£1,058*	-	• Poor	2 High 3 Medium	Remove

* Observed period only (not enough data to annualise). Observed kWh covers each meter's own period, so periods differ. Base kW is the all-period 5th-percentile reference.

Ameresco UK

Internal use. Advanced version 2.1, October 2026

For engineers, project development and sales teams

Contents

1	Quick start	From files to a client-ready PDF
2	What the tool is, and what it isn't	Scope, privacy, how much to trust it
3	Loading meter data	Formats, clock changes, duplicates, gaps
4	Data review	How the file was read, exclusions, held-back days
5	Operating calendar	Hours, holidays, closures, exceptions
6	Site settings	Rates and time bands, carbon, analysis choices
7	The portfolio view	Where to start, Opportunities, Estate comparison
8	Findings and opportunities	Impact, confidence, status, each finding
9	Base load and closed hours	Two references, monthly base, before and after
10	Heatmap and days	Whole period, week, day inspector, comparisons
11	Unusual days	Day-level anomalies and persistent changes
12	Savings scenarios	Four scenarios, no double counting
13	Capacity	Peaks, kW and kVA, headroom, test loads
14	Outputs and projects	PDFs, Excel, saving and reopening work
15	Customer editions	Password-protected dashboards for clients
16	Calculation reference	Every formula and threshold in one place
17	Limits and troubleshooting	Known limits, common problems, fixes

If you only read one page

Read section 1. Then, before you send anything to a client, check three things on every site: the **calendar** is right and ticked as confirmed, the **Data review** tab has nothing left held back that matters, and the **data quality** verdict isn't Poor. Most wrong answers come from one of those.

Standard and Advanced

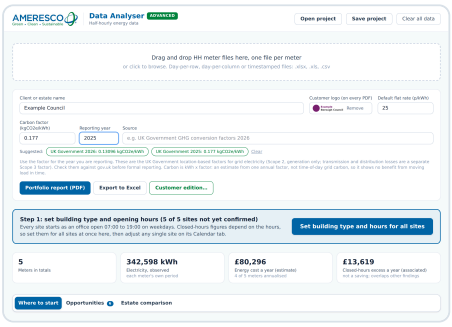
There are two versions, built from the same engine, so the same file gives the same numbers in both. **Standard** (`Ameresco-Data-Analyser-Standard.html` , grey badge) is for quick screening: Overview, Data review, Calendar, Heatmap, Capacity, Charts and Settings, with findings and reports. **Advanced** (`Ameresco-Data-Analyser-Advanced.html` , green badge) adds Base load, Unusual days, Savings scenarios, the Opportunities and Estate comparison views, opportunity tracking, project files, the week-by-week explorer and the carbon panel. This guide covers Advanced; everything in Standard works the same way. The badge shows next to the name, on the customer password page and in every PDF footer.

What changed in the Advanced version

Missing readings are no longer counted as zero. Annual figures are only shown with enough data. Calendars have per-day hours, overnight shifts, closures and exceptions. Findings carry separate impact and confidence and can be tracked as opportunities. New tabs for data review, base load, heatmap, unusual days, savings scenarios and capacity. Time-band tariffs, an optional carbon factor, estate comparison, and project files you can save and reopen. Expect some figures to differ from version 1, mainly where files had gaps.

1. Quick start

1. **Open the app.** Double-click `Ameresco-Data-Analyser-Advanced.html`. It opens in Chrome or Edge. Nothing installs and nothing goes online.
2. **Load the meter files.** Drag them onto the upload box. One file per meter; each becomes a site. Loading the same file twice is refused.
3. **Set the calendars.** Every site starts as Office / commercial with preset hours marked *unconfirmed*. Use the blue *Set building type and hours for all sites* button above the totals (it shows how many sites are still unconfirmed), then open each site's **Calendar** tab, correct the hours, add closures (school holidays, shutdowns) and tick *Hours confirmed with the site*.
4. **Check the data.** On each site's **Data review** tab, read how the file was read, decide on any held-back days (clock changes, duplicate rows) and exclude anything you know is wrong.
5. **Set the rate.** A flat default rate applies everywhere (25p/kWh). Override it on a site, or enter time bands. Enter a carbon factor if you want carbon figures; there is no default.
6. **Read the results.** *Where to start* ranks the sites. *Opportunities* lists every finding across the estate. Click a site for its tabs.
7. **Save and export.** *Save project* keeps everything (data, settings, statuses, scenarios) in one file to reopen later. Then export PDFs or Excel, or make a customer edition.



The main page: upload, client name and logo, default rate, carbon factor, exports, then the estate totals and the three views.

Before anything goes to a client

- Every calendar is checked and confirmed. Unconfirmed hours cap closed-hours findings at Medium confidence.
- The data quality verdict is Good or Check, and you've read the Check items. Don't quote figures from a Poor file.
- Annual figures say *indicative* under 330 days of valid readings and don't appear at all under 180 (unless you forced them, which shows a warning).
- Costs are **associated costs**: what a pattern costs now at the rates entered. Not savings. Scenario figures are what-ifs. Keep your covering email consistent.

2. What the tool is, and what it isn't

The Data Analyser turns half-hourly (HH) electricity data into an evidence-based list of where a site or estate uses energy when it shouldn't, priced at your rates, with a check on spare supply capacity. It handles one site or hundreds and produces Ameresco-branded PDFs and spreadsheets.

It is

- **A screening and investigation tool.** It shows where to look, how much it matters and how good the evidence is.
- **Offline and private.** Everything runs in your browser. Files are read on your laptop and never uploaded.
- **Transparent.** Every finding has a *Show calculation and evidence* link with the method, sample sizes and dates. The formulas are in section 16.

It isn't

- **An equipment diagnosis.** Whole-site HH data shows *when* demand happens, not *which* plant causes it. Checks name possible explanations to confirm on site.
- **Verified savings.** Associated costs and scenario figures are estimates. Measurement and verification needs a proper baseline and adjustment.
- **Weather normalised.** Comparisons between periods are of measured demand. A cold winter will show as more use.
- **Live monitoring.** It analyses the files you load, nothing more.
- **A connection study or bill validation.** Headroom is from the peaks in the data. Fixed and capacity charges aren't modelled.

Where the data goes

Nowhere. Closing the tab clears it; *Clear all data* does the same without closing. Project files, PDFs, spreadsheets and customer editions you export are ordinary files in Downloads, so look after them in line with the client's data agreement. Project files contain the half-hourly data and your internal notes.

Browser and versions

Use current **Chrome or Edge**. The app is one file of about 2.3 MB and doesn't update itself; replace your copy when a new version comes out. The Standard and Advanced versions are separate files and can sit side by side. Project files record their format version, so a later version of the app can tell what it is reading.

3. Loading meter data

Files and layouts

- **File types:** `.xlsx`, `.xlsm`, `.xls`, `.csv`, `.txt`. Comma, semicolon or tab separated.
- **One meter per file.** The file name becomes the site name. A meter reference (MPAN or serial) is picked up if the file has one, and a warning shows if two files carry the same meter.

Layout	Looks like	Notes
Days down the side	A date, then 48 half-hourly readings.	Meter columns before the date are ignored.
Days across the top	One column per day, 48 rows down.	Turned round automatically.
46 or 50 periods	Exports that follow clock changes.	Spring days have 46 readings, autumn days 50. Both handled exactly.
Timestamped	One row per reading: date-time and value.	15-minute data is added up to half hours. Hourly data is refused. UTC timestamps are converted to UK time.

How readings are treated

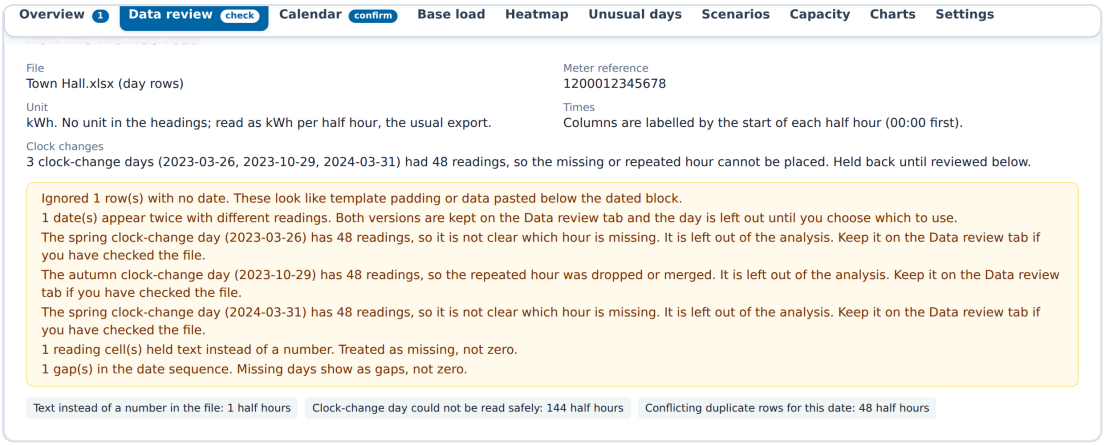
- **Missing is not zero.** Blank cells, text (error codes) and absent days are *missing*. They're left out of every figure and shown hatched on the heatmap. Annual figures scale from valid hours only.
- **Units.** Readings are taken as kWh per half hour unless the headings say kW, in which case they're converted. Check the unit on the Data review tab.
- **Start or end labels.** Whether a column labelled 00:30 means the half hour starting or ending then is detected from the headings and shown. Change it in Settings if the supplier says otherwise.
- **Clock changes.** A spring day with 46 readings and an autumn day with 50 are read correctly. A clock-change day with 48 readings can't be placed safely, so it's *held back* until you review it.
- **Duplicate rows.** Identical repeats are counted once. Two different rows for the same date are both kept and the day is held back until you choose which to use.
- **Negative readings** are a fault on an import meter, but normal on export or net meters. Set the meter type in Settings.

What it refuses

A file with **several meters stacked** one after another (lots of conflicting repeated dates) is refused rather than spliced into one profile. Split it into one file per MPAN.

4. Data review

Each site's **Data review** tab shows what the app did with the file and lets you correct it. Nothing you do here changes the original file; it's all reversible.



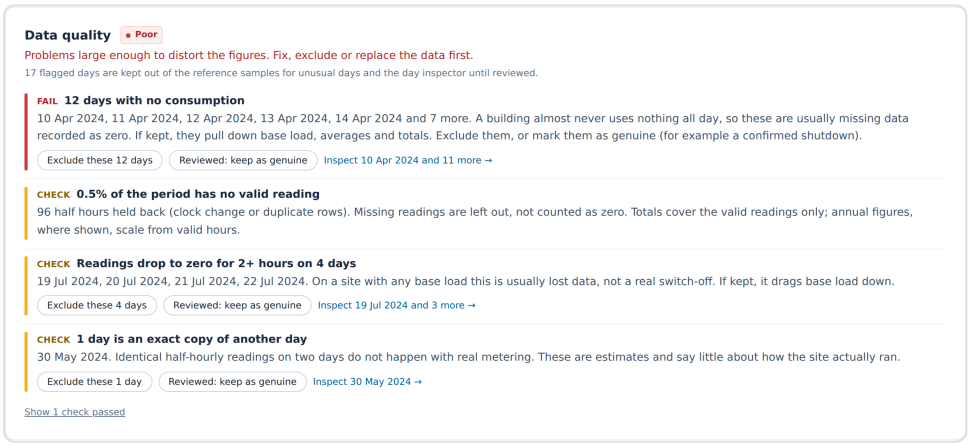
How the file was read: layout, meter reference, unit, time convention, clock changes and data notes, with counts of flagged half hours.

Held back for review

Clock-change days that couldn't be placed and dates with conflicting duplicate rows. For each, keep it (or pick the first or second row) or leave it out. Until you decide, it's left out.

Exclusions

Leave out readings you know are wrong: a meter fault, a period of works, estimated data. Give a date range, optionally a time window, and a reason. Excluded readings are left out of every figure, shown pink on the heatmap, and listed in the Excel and PDF appendix.



The faulty Library meter: the data quality check found days of zero readings. One click excludes them with the reason recorded.

Data quality check

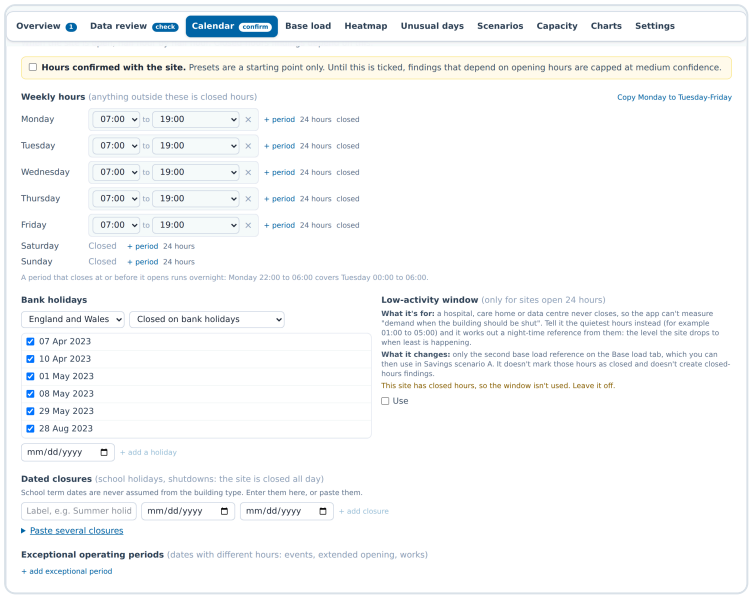
Every file is checked and given a verdict: **Good**, **Check** (usable, read the flagged items) or **Poor** (fix the data before quoting figures). Each check lists its dates with buttons to exclude them, mark them as fine, or inspect them.

Check	What it looks for	Check / Fail
Coverage	Days' worth of valid readings.	Check under 330; Fail under 180
Missing readings	Absent days, blank or text cells, held-back half hours.	Fail if over 10% of the period
Zero days	Whole days with (almost) no use: usually missing data recorded as zero.	Fail if over 3% of days
Dropouts	Exactly zero for 2 hours or more inside a normal day.	Fail if over 5% of days
Negative readings	On an import meter only.	Always Fail
Stuck readings	The same non-zero value for 6 hours or more (estimates).	Fail if over 10% of days
Copied days	A day identical to another.	Fail if over 5% of days
Spikes	Over 3× the 99.5th percentile and 5 kW above it.	Check only

Flagged days not yet reviewed are kept out of reference samples (comparable days, unusual-day baselines), so one bad day can't distort the comparison.

5. Operating calendar

The calendar says when the site is open, half hour by half hour. Every closed-hours figure depends on it, so it's the most important setting.



The Calendar tab: confirmation, weekly hours, bank holidays, low-activity window, dated closures and exceptional periods. Date boxes follow your computer's date format.

- **Weekly hours.** Each day has its own hours and can have several periods (e.g. 07:00 to 12:00 and 13:00 to 19:00). A period that closes at or before it opens runs overnight: Monday 22:00 to 06:00 covers Tuesday morning.
- **Bank holidays.** Choose England and Wales, Scotland or Northern Ireland, then closed, Sunday hours or normal hours. Untick a date to treat it as a normal day, or add a local holiday.
- **Dated closures.** School holidays, Christmas shutdowns: the site is closed all day. School term dates are never assumed from the building type; enter them here or paste several at once (one per line, e.g. `24/12/2025 - 02/01/2026 Christmas`).
- **Exceptional periods.** Dates with different hours: events, extended opening, works.
- **Low-activity window (24-hour sites only).** A hospital or data centre never closes, so there is no "demand when shut" to measure. Enter its quietest hours (say 01:00 to 05:00) and the app works out a night-time reference from them, shown as the second reference on the Base load tab and usable in Savings scenario A. It doesn't mark those hours as closed or create closed-hours findings. Leave it off for sites that close.
- **Hours confirmed.** Building-type presets are a starting point. Until you tick this, calendar-dependent findings are capped at Medium confidence and the site shows *unconfirmed*.

The bulk setting on the main page replaces every site's weekly hours and holiday rules, and adds any closures entered there to each site's own.

6. Site settings

Rates and time bands

A flat rate in p/kWh, fully delivered. Or switch to **time bands**: each band has days, a time window and a rate (e.g. weekday day, night, weekend). The app checks that every half hour has exactly one band; if not, it lists the problem and uses the flat rate until it's fixed. Bank holidays can be priced as Sundays. Fixed and capacity charges aren't included and the reports say so.



Time bands with the validation message.

Carbon

Enter a factor in kgCO2e/kWh, the reporting year and the source on the main page, or click one of the suggested UK Government factors to fill all three: **2026: 0.13096** and **2025: 0.177** kgCO2e/kWh (location-based Scope 2, generation only; transmission and distribution losses are a separate Scope 3 factor). Use the factor for the year you are reporting, and check it against gov.uk before formal reporting. Nothing is applied by default, so no carbon is shown until a factor is entered. It's kWh × one annual factor: it doesn't show any benefit from moving load in time.

Site details

- **Floor area (m²)** switches on kWh/m² in the estate comparison.
- **Physical site and role.** Meters with the same physical site name are grouped. Mark each as main, submeter or separate. Submeters are never added to their main meter, in the site group or in estate totals.
- **Meter type** (import, export, net) decides whether negative readings are a fault.

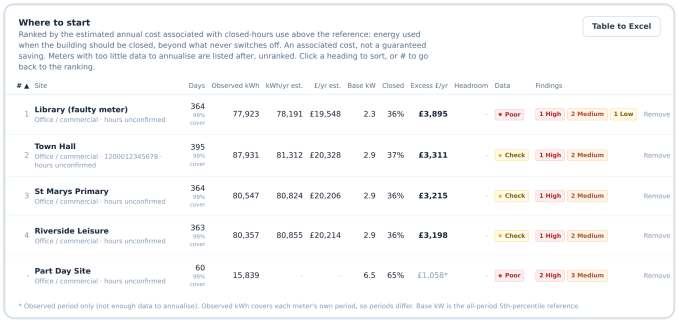
Analysis choices

Setting	What it does
Annual estimates	*Automatic*: full at 330+ valid days, indicative from 180, off below. *Force* annualises short data with a warning on every figure. *Off* shows observed figures only.
Excess reference	Whether closed-hours excess is measured above the all-period reference (default) or the closed-hours reference (section 9).
Base percentile	Which percentile sets the base reference. Default 5th.
Unusual-day thresholds	Comparable days, minimum days, % and MAD thresholds, and the persistent-change rule (section 11).

7. The portfolio view

The main page shows estate totals (observed kWh over each meter's own period, annual cost for meters that can be annualised, submeters excluded), then three views.

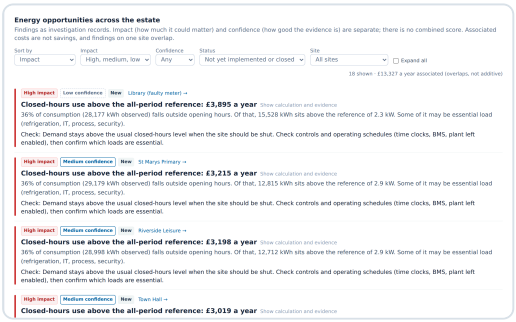
Where to start



Ranked by the annual cost associated with closed-hours excess. Click a heading to sort; Table to Excel saves it in the order shown.

Sites are ranked by the **associated annual cost of closed-hours use above the reference**: energy used when the building should be shut, beyond what never switches off. It's usually the cheapest thing to tackle, so it's a good first sort. Meters with too little data to annualise are listed after, unranked. Columns show valid days and coverage, observed kWh, annual kWh and cost (estimates), base kW, closed-hours share, the ranking figure, firm headroom, the data verdict and finding counts.

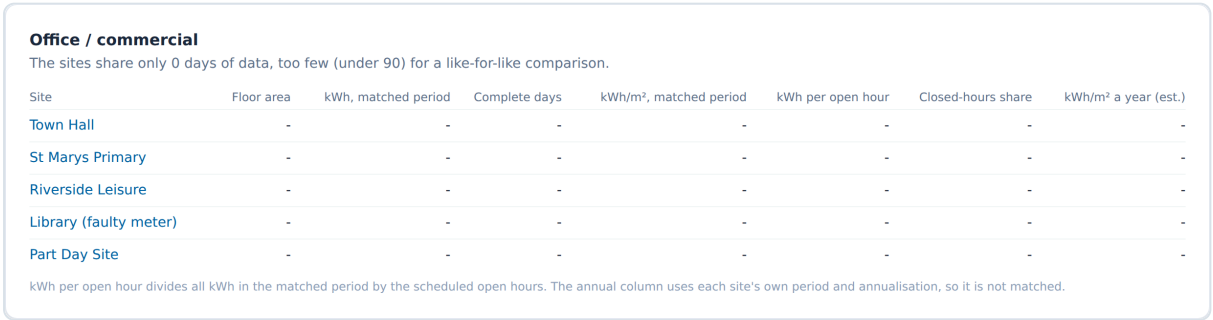
Opportunities



Every finding across the estate as a record, filterable by impact, confidence, status and site.

Use it as the action list for a client: sort by impact or confidence, filter to what's not yet done, and set statuses as you go (section 8). The total shown is not additive, because findings on one site overlap.

Estate comparison



Office / commercial

The sites share only 0 days of data, too few (under 90) for a like-for-like comparison.

Site	Floor area	kWh, matched period	Complete days	kWh/m², matched period	kWh per open hour	Closed-hours share	kWh/m² a year (est.)
Town Hall	-	-	-	-	-	-	-
St Marys Primary	-	-	-	-	-	-	-
Riverside Leisure	-	-	-	-	-	-	-
Library (faulty meter)	-	-	-	-	-	-	-
Part Day Site	-	-	-	-	-	-	-

kWh per open hour divides all kWh in the matched period by the scheduled open hours. The annual column uses each site's own period and annualisation, so it is not matched.

Sites compared only with sites of the same type, over the dates they all cover.

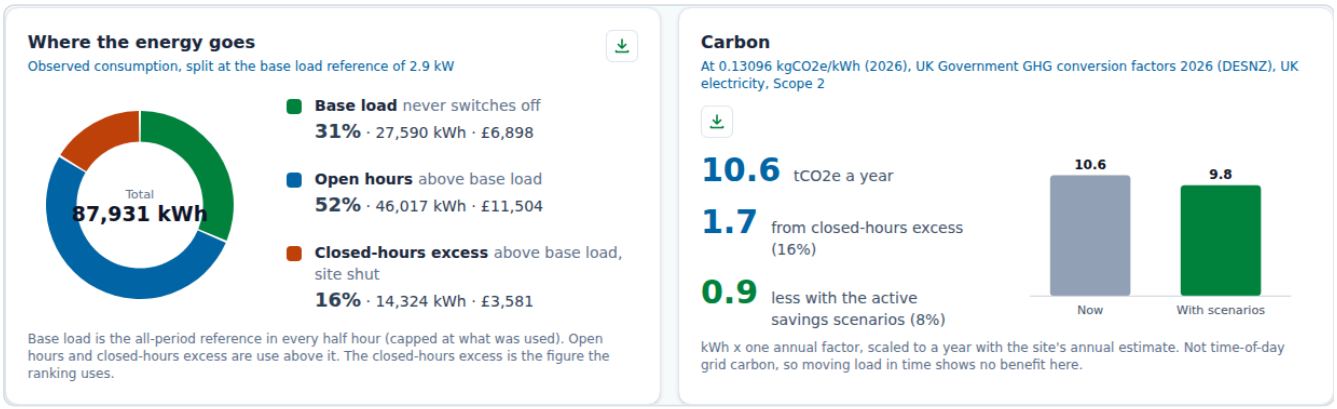
Raw consumption doesn't show which site is better run, so the comparison is fair or it isn't shown: same building type, at least 90 shared days, complete days only, and kWh/m² where floor areas are entered, plus kWh per scheduled open hour. Nothing is labelled best or worst. The sample sites cover different years, so they share no days, which is why the table above is empty.

8. Findings and opportunities

Each site has ten tabs on a bar that stays pinned to the top of the window as you scroll, so you can always jump to another one. The active tab is filled blue; a small badge flags something to look at (high findings, a data check, a calendar to confirm, the number of unusual days or active scenarios).



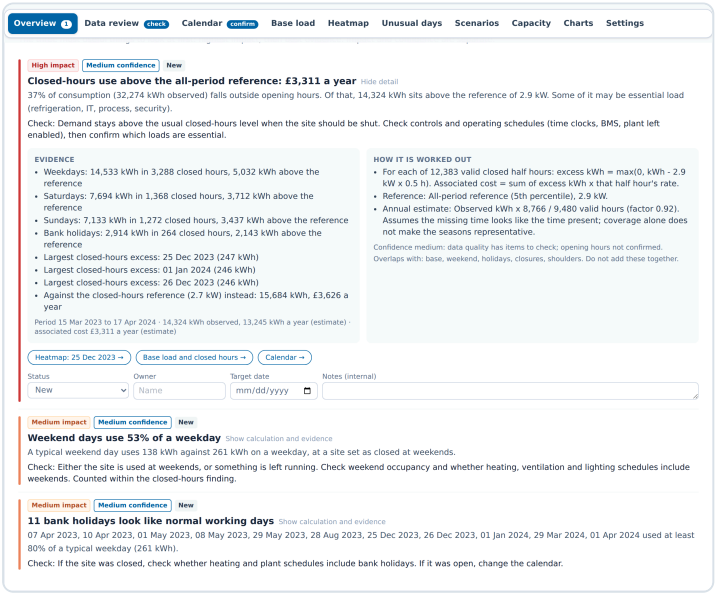
The site tabs.



Top of the Overview: where the energy goes, and the carbon panel (Advanced, once a factor is entered).

Where the energy goes splits observed consumption three ways, which add up to the total: **base load** (up to the reference in every half hour: what never switches off), **open hours** above it, and the **closed-hours excess** above it, which is the ranking figure. It also appears in the PDF appendix. The **carbon panel** shows tCO2e at the entered factor, the share from closed-hours excess and what the active savings scenarios would remove. Every chart has a download button (green arrow) that saves it as a picture for proposals.

A site's **Overview** tab shows its key figures (each tagged *Measured* or *Estimate*), its **top three investigations** and all findings.



Top investigations, with the calculation and the opportunity record open on the first.

Impact and confidence are separate

- **Impact** is how much it could matter: High at 10% or more of the site's energy cost, Medium at 4%, Low below. Some findings have a fixed level.
- **Confidence** is how good the evidence is. It starts High and is capped by: Poor data (Low) or Check data (Medium); under 180 valid days (Low) or under 330 (Medium); unconfirmed hours (Medium, for calendar-dependent findings); a small sample. The reasons are listed in the evidence.

There's no combined score. The top three are the highest impact, then the best evidence.

Tracking

Each finding is an opportunity record with a status (New, Under review, Site visit required, Confirmed, Action agreed, Implemented, Closed), an owner, a date and notes. These save with the project. Customer editions show status but leave out owner and notes unless you choose to include them.

Finding	When it appears	Level
Closed-hours use above the reference	Any site with closed hours. Closed-hours energy above the reference, priced.	By cost
Load that never switches off	Always. Base reference, its share and cost. Each 1 kW is 8,766 kWh a year.	Medium if half or more of use at a site that closes
Weekend running	Site closed at weekends, but a typical weekend day uses 50%+ of a weekday.	High at 70%+
Bank holidays like working days	A holiday uses 80%+ of a typical weekday at a site closed on them.	Medium
Closures running	Weekdays in a dated closure use 60%+ of a normal weekday.	Medium
Early starts, late stops	Closed-hours excess in the two hours either side of opening.	By cost
Base load change	Base up or down 15%+ on the same months a year earlier (needs about 14 months).	By cost if up
Unusual days	Days far from their matched baseline (section 11).	Low
Persistent change	A sustained step up or down (section 11).	Medium if up
Supply capacity	Peak at 90%+ of capacity, or 50% or less.	High at 90%+

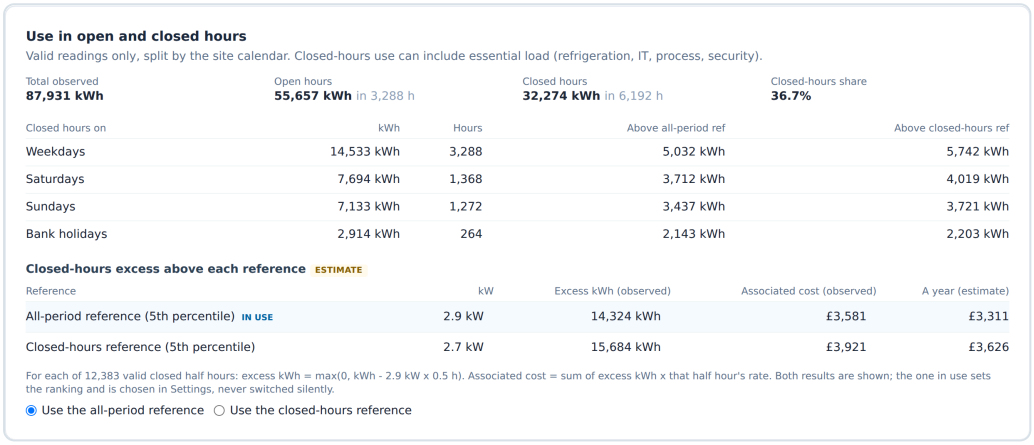
> **How to use the £ figures**
>
> They're associated costs at the rates entered, not savings, and they overlap (weekend running is inside the closed-hours figure), so don't add them. For a what-if, use Savings scenarios, which don't double count.

9. Base load and closed hours

The app shows **two references** and says which one each figure uses.

- **All-period reference:** the 5th percentile of every valid half hour. The load that is almost always there. Shown next to the lowest single half hour so you can see they differ.
- **Closed-hours reference:** the 5th percentile of valid closed half hours only, with the sample size. Better for sites that run plant overnight on purpose.

Closed-hours excess for each valid closed half hour is `max(0, kWh − reference kW × 0.5 h)`, priced at that half hour's rate, summed, then scaled to a year if the site is annualised.



Use in open and closed hours

Valid readings only, split by the site calendar. Closed-hours use can include essential load (refrigeration, IT, process, security).

Total observed	Open hours	Closed hours	Closed-hours share
87,931 kWh	**55,657 kWh** in 3,288 h	**32,274 kWh** in 6,192 h	**36.7%**

Closed hours on	kWh	Hours	Above all-period ref	Above closed-hours ref
Weekdays	14,533 kWh	3,288	5,032 kWh	5,742 kWh
Saturdays	7,694 kWh	1,368	3,712 kWh	4,019 kWh
Sundays	7,133 kWh	1,272	3,437 kWh	3,721 kWh
Bank holidays	2,914 kWh	264	2,143 kWh	2,203 kWh

Closed-hours excess above each reference ESTIMATE

Reference	kW	Excess kWh (observed)	Associated cost (observed)	A year (estimate)
All-period reference (5th percentile) IN USE	2.9 kW	14,324 kWh	£3,581	£3,311
Closed-hours reference (5th percentile)	2.7 kW	15,684 kWh	£3,921	£3,626

For each of 12,383 valid closed half hours: excess kWh = max(0, kWh - 2.9 kW x 0.5 h). Associated cost = sum of excess kWh x that half hour's rate. Both results are shown; the one in use sets the ranking and is chosen in Settings, never switched silently.

◉ Use the all-period reference ○ Use the closed-hours reference

Open and closed use by day class (weekdays, weekends, bank holidays, closures, exceptions), with excess against both references.



Base load by month, with the number of valid half hours and excluded readings behind each bar.

Did a change make a difference? (before and after)

Use it when you know something changed on site on a date: a new BMS or time-clock schedule, plant switched off, an LED or controls upgrade. Enter the date and a window (2 to 12 weeks, 8 by default). The app compares the weeks before with the same number of weeks after: average demand, base load and average demand when closed. It then says in one line what moved and what it would be worth over a year at the site's average rate if it held.

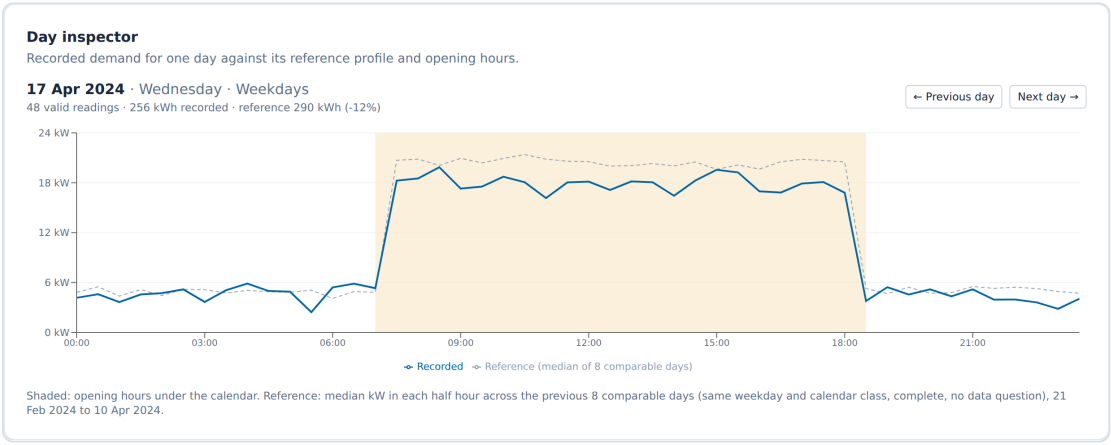
Keep the window short so the seasons don't swamp the answer: a change made in November will look like a rise just because winter started. It shows what changed in the meter data, not why.

10. Heatmap and days



The whole period. Hatched cells have no valid reading, pink cells are excluded or held back, orange marks show opening and closing times.

The fastest way to read a site. Look for dark bands outside the orange marks, weekends that look like weekdays, holidays that don't show, step changes and gaps. Hover for the reading; click a day to inspect it.



The day inspector: the day against its reference profile, with opening hours shaded.

The **reference profile** is the median kW in each half hour across the previous 8 comparable days (same weekday and calendar class, complete, no unresolved data question). The number of days and their range are shown. Below it, the **week view** shows seven days as a grid.

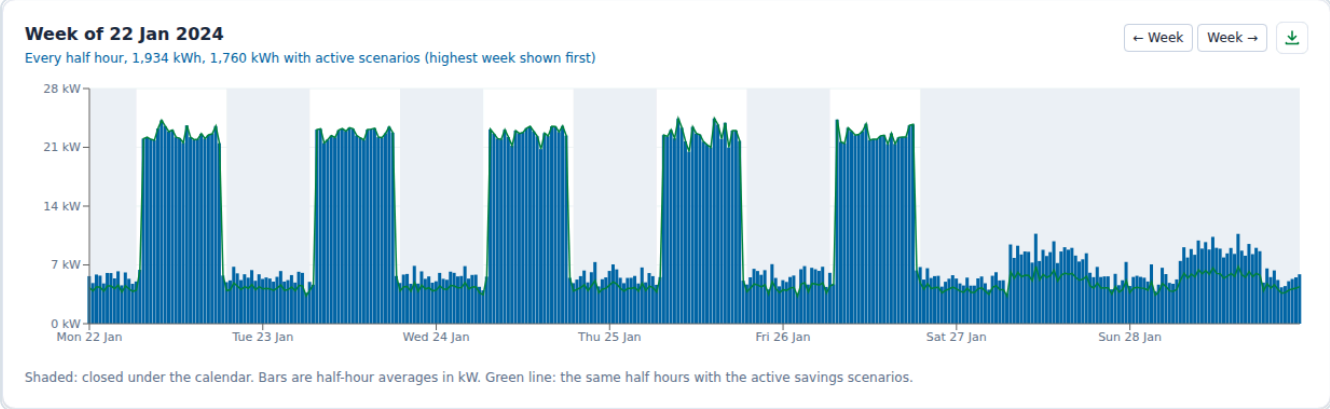
Compare periods

Answers "is the site using more or less than before?". Pick a period (quick picks: last 4 weeks, 3 months, 6 months) and the app compares it with the period just before it and with the same weeks a year earlier, in one sentence each. Use it to show a customer how a recent month compares, to check that savings are holding, or to spot a site creeping up. The year-earlier comparison is the fairer one because it is the same season. Only days complete in both periods are counted; if too few match, it says so rather than guess.

Week by week (Advanced)



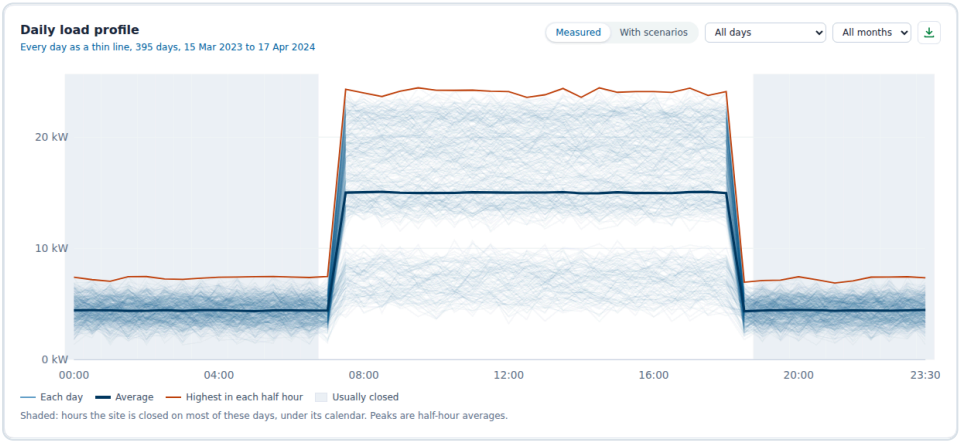
Weekly totals; click a week to open it. Pale bars have under 90% of the week's readings.



One week, half hour by half hour, with closed hours shaded and the active scenarios as a green line.

Below the heatmap, a bar per week (kWh or £). Click any bar to see that week half hour by half hour, with closed hours shaded, the result of active savings scenarios, and, when time bands are set, the unit rate in each half hour underneath. It opens on the highest week.

Daily load profile (Charts tab)



Every day as a faint line, the average in dark blue, the highest in each half hour in orange.

The quickest way to see how consistent a site is: where the lines bunch, the site behaves the same every day; a second band means a different kind of day (here, weekends); stray lines are one-offs. Filter by days (all, open, closed, weekdays, weekends) and month. In Advanced, switch to *With scenarios* to see the effect of the active savings scenarios against the measured average.

The Charts tab keeps the earlier analyser charts (day-of-week profile, load duration curve, day and night by month, week comparison, every reading). Its date filter affects those charts only; findings always use the full record.

11. Unusual days

Unusual days — Largest first | By date

Date	Compared with	Actual	Expected	Difference	Baseline	Data	
10 Apr 2024	Wednesday	0 kWh	290 kWh	-100%	8 days, 14 Feb 2024 to 03 Apr 2024	no consumption all day	Inspect →
11 Apr 2024	Thursday	0 kWh	294 kWh	-100%	8 days, 15 Feb 2024 to 04 Apr 2024	no consumption all day	Inspect →
12 Apr 2024	Friday	0 kWh	303 kWh	-100%	8 days, 09 Feb 2024 to 05 Apr 2024	no consumption all day	Inspect →
13 Apr 2024	Saturday	0 kWh	158 kWh	-100%	8 days, 17 Feb 2024 to 06 Apr 2024	no consumption all day	Inspect →
14 Apr 2024	Sunday	0 kWh	158 kWh	-100%	8 days, 11 Feb 2024 to 07 Apr 2024	no consumption all day	Inspect →
15 Apr 2024	Monday	0 kWh	297 kWh	-100%	8 days, 12 Feb 2024 to 08 Apr 2024	no consumption all day	Inspect →
16 Apr 2024	Tuesday	0 kWh	287 kWh	-100%	8 days, 20 Feb 2024 to 09 Apr 2024	no consumption all day	Inspect →
17 Apr 2024	Wednesday	0 kWh	290 kWh	-100%	8 days, 14 Feb 2024 to 03 Apr 2024	no consumption all day	Inspect →
18 Apr 2024	Thursday	0 kWh	294 kWh	-100%	8 days, 15 Feb 2024 to 04 Apr 2024	no consumption all day	Inspect →
19 Apr 2024	Friday	0 kWh	303 kWh	-100%	8 days, 09 Feb 2024 to 05 Apr 2024	no consumption all day	Inspect →
20 Apr 2024	Saturday	0 kWh	158 kWh	-100%	8 days, 17 Feb 2024 to 06 Apr 2024	no consumption all day	Inspect →
21 Apr 2024	Sunday	0 kWh	158 kWh	-100%	8 days, 11 Feb 2024 to 07 Apr 2024	no consumption all day	Inspect →

Possible checks: site events, works, faults, occupancy changes, or missing data recorded as low readings.

Unusual days at the faulty Library meter, each against its own baseline. Here the data column shows they're meter faults, not operations.

Each day is compared with the **previous 8 comparable days**: same actual weekday and the same calendar class (so a bank holiday Monday is compared with bank holidays, not Mondays). At least 6 are needed; days without enough are counted as *cannot apply*. Days with unresolved data problems are kept out of the baseline.

A day is flagged when it's more than **25%** from the baseline median **and** more than **4 × the scaled median absolute deviation**. High and low days are both flagged. A low day is not a saving until the data and an operational change are confirmed.

Persistent changes

A sustained shift rather than a one-off: demand at least **15%** above or below its baseline on **5 of 7** consecutive eligible days. Reported once with the date it started.

Persistent changes

Demand at least 15% above or below its matched baseline on 5 of 7 consecutive eligible days.

From	To	Direction	Days flagged	Average	Actual vs expected	
08 Apr 2024	29 May 2024	Below baseline	43 of 50	-38%	7,684 kWh vs 12,349 kWh	Inspect →

A change in measured demand only. Check what changed on site; a fall is not a saving until the data and an operational change are confirmed.

Persistent changes at the Library.

All these thresholds can be changed in the site's Settings.

12. Savings scenarios

What-if estimates applied to **one copy** of the half-hourly data, in the order listed. Each scenario works on what earlier ones left, so nothing is counted twice, and no half hour goes below its retained floor or below zero. Every input starts at zero.

Scenario	What it does
A. Reduce closed-hours excess	Removes a share of demand above the reference in every closed half hour. The reference (the load that never switches off) stays, so even 100% leaves that level running overnight and at weekends. Add B to test cutting it.
B. Reduce base load in chosen hours	Removes a fixed kW in a time window on chosen days, never below the floor.
C. Remove early-start or late-stop demand	Removes a share of demand above a floor in windows around opening and closing.
D. Lower peak demand	Usually *Hold demand at or below* a target kW: every half hour above it is brought down to it. One click sets the target to 90%, 80% or 70% of the current peak, and the result shows how many half hours on how many days that touches. You can also trim the N highest half hours or every half hour above a level. *Remove* takes the energy away; *shift* moves it to a chosen window the same day. Expect a small energy saving: the value of lower peaks is in capacity and peak-time network charges (not priced here) and electrification headroom.



Results for scenarios A and B together: removed kWh and cost in the observed period, an annual projection, carbon if a factor is entered, and the average day before and after.

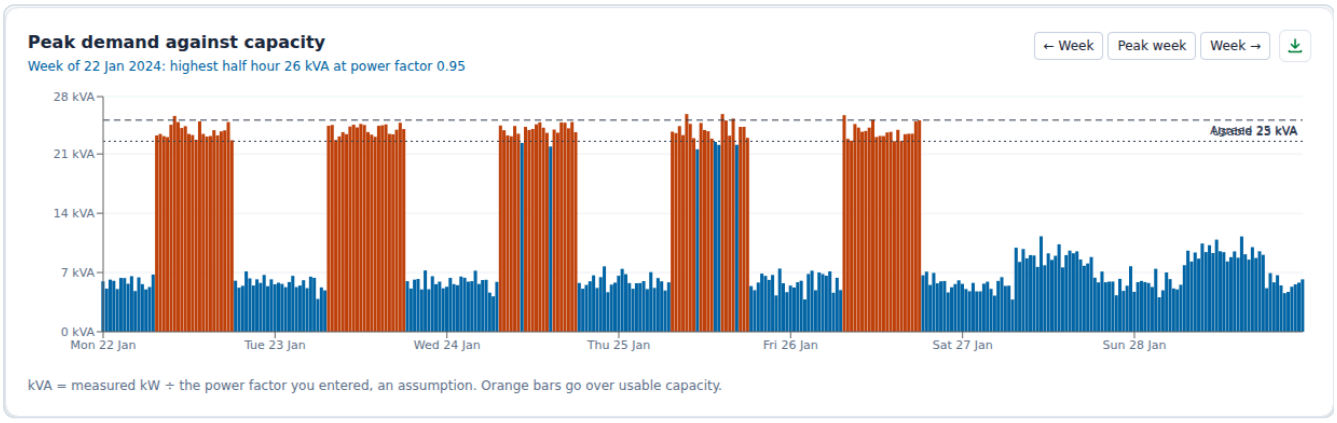
The chart shows the average day before and after. Switch between **all days**, **days the site opens** and **closed days** (weekends, holidays, closures). "All days" mixes the two, so daytime still shows working-day use even when closed-hours demand is cut; "closed days" shows scenario A clearly, with demand brought down to the dashed reference line.

> **Check figure**
>
> 10 kW removed for 12 hours a day on 100 days is 12,000 kWh. At 25p/kWh that's £3,000. The app is tested against this.

Scenarios save with the project and appear in the Excel export (with an interval sheet for internal use) and the PDF appendix. Customer editions show them read-only.

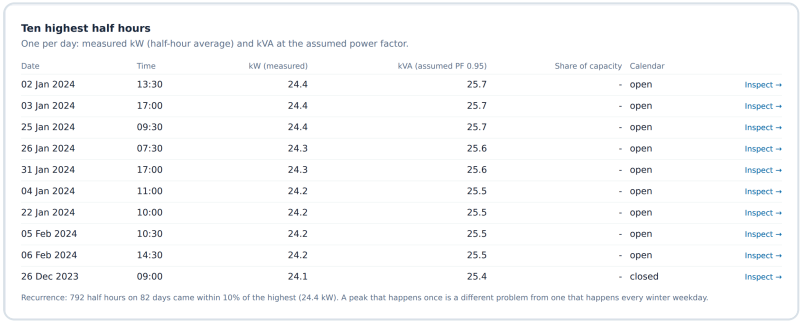
13. Capacity

Enter the agreed supply capacity (kVA), power factor and safety margin on the **Capacity** tab.



Peak demand against capacity: one week in kVA, with the agreed and usable capacity (and a peak target from scenario D). Orange bars are over the line.

The chart opens on the week of the highest half hour; step through weeks with the arrows. Then the ten highest half hours, one per day:



Ten highest half hours

One per day: measured kW (half-hour average) and kVA at the assumed power factor.

Date	Time	kW (measured)	kVA (assumed PF 0.95)	Share of capacity	Calendar	
02 Jan 2024	13:30	24.4	25.7	-	open	Inspect →
03 Jan 2024	17:00	24.4	25.7	-	open	Inspect →
25 Jan 2024	09:30	24.4	25.7	-	open	Inspect →
26 Jan 2024	07:30	24.3	25.6	-	open	Inspect →
31 Jan 2024	17:00	24.3	25.6	-	open	Inspect →
04 Jan 2024	11:00	24.2	25.5	-	open	Inspect →
22 Jan 2024	10:00	24.2	25.5	-	open	Inspect →
05 Feb 2024	10:30	24.2	25.5	-	open	Inspect →
06 Feb 2024	14:30	24.2	25.5	-	open	Inspect →
26 Dec 2023	09:00	24.1	25.4	-	closed	Inspect →

Recurrence: 792 half hours on 82 days came within 10% of the highest (24.4 kW). A peak that happens once is a different problem from one that happens every winter weekday.

The ten highest half hours, one per day, in kW and kVA.

- **kW vs kVA.** Half-hourly data gives kW (a half-hour average, not an instantaneous maximum). kVA = kW ÷ power factor, which is an assumption you enter (0.95 if unknown).
- **Recurrence.** How many half hours, on how many days, came within 10% of the peak. One-off peaks are easier to manage than recurring ones.
- **Firm headroom** = (usable kVA − peak kVA) × power factor, where usable kVA is the agreed capacity less the margin (10% default).
- **Test a new load** (heat pumps, EV charging): enter kW, hours and season. The result says whether it would exceed usable capacity in the analysed half hours under the entered assumptions.

> **Wording**
>
> "No exceedance" means none in the analysed half hours under the entered assumptions. A colder year, load on other meters or DNO constraints can change the answer. It's a screen before a capacity conversation, not a substitute.

14. Outputs and projects

Projects

Save project writes one `.adproject` file with the raw readings, data review decisions, calendars, rates, statuses, notes, scenarios, client name, logo and carbon factor. **Open project** brings it all back. Use it to pick work up later or hand it to a colleague. It holds client data and internal notes, so treat it like the source files.

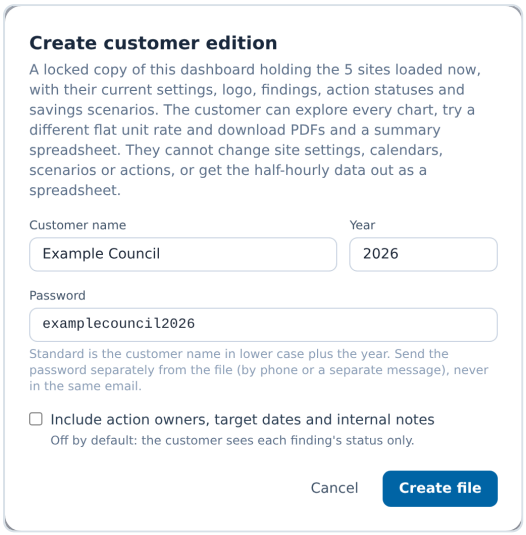
Exports

Output	What's in it	Audience
Site summary (PDF)	One page per site, plus an appendix: top three, every finding with its calculation, data review, calendar and tariff.	Client-ready
Download all charts (PDF)	Full site report with every chart and table, plus the appendix.	Client-ready
Charts as separate PDFs (zip)	The same, one PDF per chart.	Client-ready
Portfolio report (PDF)	Ranked table, estate opportunities, then a page per site.	Client-ready
Table to Excel	The Where to start table in the order shown.	Internal or client
Export to Excel	Summary, Opportunities, Scenarios, Scenario intervals, Data review, Calendar and tariff, and a sheet per site with the half-hourly readings (missing left blank, with a flag column).	Internal (raw data and notes)
Customer edition	A locked dashboard for one client (section 15).	Client

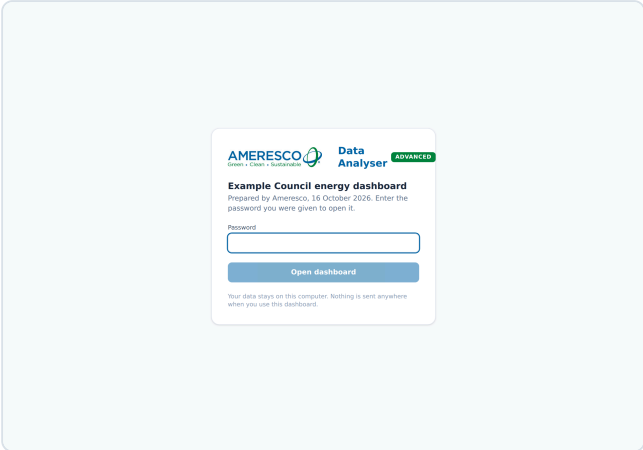
Internal notes and owners appear only in internal exports. Set the client name and logo before exporting.

15. Customer editions

A single HTML file holding one client's sites, locked with a password. The client explores their data, adjusts the things they know better than we do, and downloads their own reports.



Creating an edition. Internal notes are left out unless you tick the box.

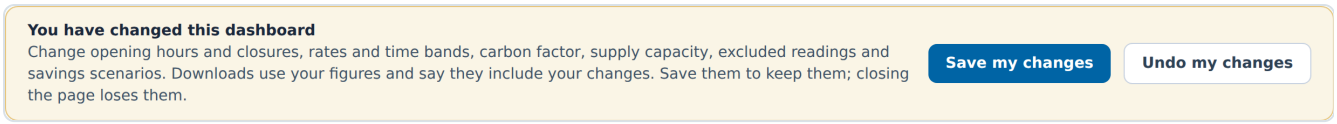


What the client sees first.

1. Load only that client's sites (or open their project), set them up fully, add the client name and logo.
2. Click **Customer edition…**, check the name and year, and decide whether to include internal notes (normally not).
3. Click **Create file**, open it yourself with the password, and check it.
4. Send the file and the password **separately**.

> **The standard password is guessable**
>
> It's the customer name in lower case, letters and digits only, then the year (e.g. `examplecouncil2026`). For anything sensitive, add a random part and keep a note of it.

The client can	The client can't
Explore every tab, chart and table, and see statuses	Add or remove sites
Change opening hours, closures, holidays and confirm the calendar	Change opportunity statuses, owners or notes
Change rates (flat or time bands), carbon factor, site name, type and floor area	Change how the file was read, meter grouping or the analysis method
Set supply capacity and test loads; exclude readings they know are wrong; build savings scenarios	See owners or internal notes (unless you included them)
Download PDFs and a summary spreadsheet; save their changes as a new copy	Export the half-hourly readings



Once the client changes anything, a banner offers to save the changes or undo them.

- **Save my changes** downloads a new copy of the dashboard with their changes, locked with the same password. Closing the page without saving loses them, and the browser warns first.
- **Undo my changes** goes back to the file as it was opened.
- Every PDF made after a change carries *includes changes made by [customer]* in its footer, so a report with their assumptions is never mistaken for ours. Ask for their saved copy if you want to see what they changed.

Encrypted with AES-256-GCM, key from the password (PBKDF2-SHA-256, 310,000 rounds). An edition is a snapshot: it can't be updated or withdrawn once sent, and a forgotten password can't be recovered.

16. Calculation reference

Figure	How it's worked out
Demand (kW)	kWh in the half hour ÷ 0.5 h. A half-hour average.
Valid reading	A number, not excluded, not held back. Blank, text and absent readings are missing, never zero.
Observed kWh and cost	Sum of valid readings, each priced at its own half hour's rate.
Annualisation	Observed × 8,766 ÷ valid hours. Full at 330+ valid days, indicative from 180, off below 180 unless forced.
All-period reference	5th percentile (setting) of all valid half-hour kW.
Closed-hours reference	5th percentile of valid closed half-hour kW (low-activity window at 24-7 sites).
Closed-hours excess	Σ max(0, kWh − reference kW × 0.5) over valid closed half hours, × rate.
Base load cost	Reference kW × 0.5 h × each valid half hour's rate, summed.
Complete day	At least 90% of its half hours valid. Day comparisons use complete days only.
Impact	£ ÷ site energy cost: 10%+ High, 4%+ Medium, below Low.
Weekend, holidays, closures	Median complete weekend day ÷ median weekday (50%, High 70%); holidays at 80%+; closure weekdays at 60%+.
Base load change	Monthly 5th percentile (months with 20+ days), paired a year apart; reported at ±15%. Cost = kW change × 8,766 × average rate.
Reference profile	Median kW per half hour over the previous 8 comparable complete days.
Unusual day	Baseline = previous 8 comparable days (minimum 6). Flagged if |difference| > 25% and > 4 × 1.4826 × MAD.
Persistent change	≥15% from baseline on 5 of 7 consecutive eligible days.
Peak kVA, firm headroom	kW ÷ power factor. (Capacity × (1 − margin) − peak kVA) × power factor.
Carbon	kWh × entered factor. No default.
Estate comparison	Same type, shared complete days (90+), kWh/m² and kWh per scheduled open hour over those days.

The core maths is tested against the Python version of the analysis on reference files, and the brief's acceptance checks run as automated tests.

17. Limits and troubleshooting

Known limits

- **No fixed or capacity charges** in costs. Time bands cover unit rates only.
- **No weather normalisation**, so year-on-year and before-and-after comparisons include weather.
- **Timestamped files:** after a project is saved and reopened, cells that held text show as blank (still missing, so the figures don't change).
- **Large estates** work but slow down; load in batches if the browser struggles.

Troubleshooting

Problem	What to do
File refused: several meters stacked	Split it into one file per MPAN.
Hourly data refused	The tool needs half-hourly (or 15-minute) data. Ask the supplier for HH.
"Already loaded"	The same file is already a site. Use the existing one.
Days missing from the analysis	Look at *Held back for review* on the Data review tab: clock-change days and conflicting duplicates wait for your decision.
No annual figures	Under 180 valid days. Use the observed figures, or force annualisation in Settings and accept the warning.
Everything is Medium confidence	Confirm the calendar, and resolve the data quality Check items.
Closed-hours figure far too high	Check the calendar. A 24-7 site set as an office will show most of its use as closed hours.
Time bands ignored	Read the validation message: a gap or overlap makes the app fall back to the flat rate.
Customer edition won't open	Check the password. Save the file and open it in Chrome or Edge, not an email preview.

> **Questions, bugs and ideas**
>
> Send them to the app owner with the file that caused the problem (or a description if it's confidential), what you expected and what you saw. A saved project file is the easiest way to show the problem.